Greedy Pair-Merge: Background Theory

Bottom-up Agglomerative Graph Coarsening

- Every node starts as its own individual cluster.
- Purely greedy: no undo or backtracking once merged.
- Selects the locally optimal legal pair in every iteration.
- Widely used in CAD, mesh coarsening, and partitioning.

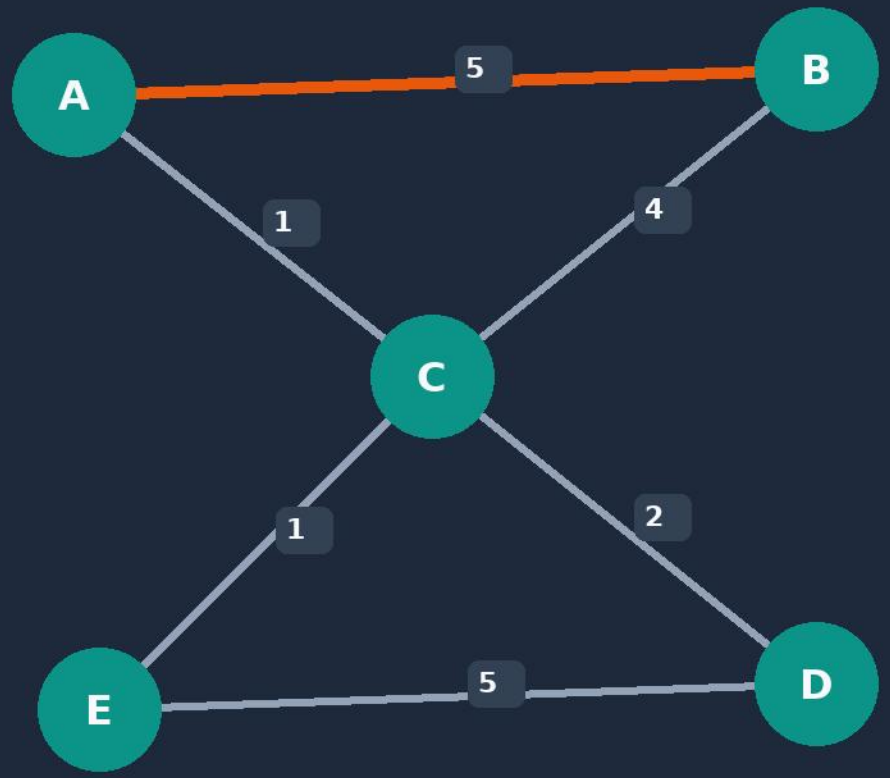
Engine Rules & Selection Mechanics

Pair Ranking, Affinity & Termination

- Initialization: singletons (every cell alone).
- Contraction: highest-affinity legal pair merges.
- Termination: reach K, or no positive legal pair remains.
- Affinity: sum of edge weights between clusters.
- Tie-break: lexicographical ordering of sorted labels.

Step 1: Every Node Starts Alone

K=5 Singletons | Target K=2 | Initial Cutsize: 18

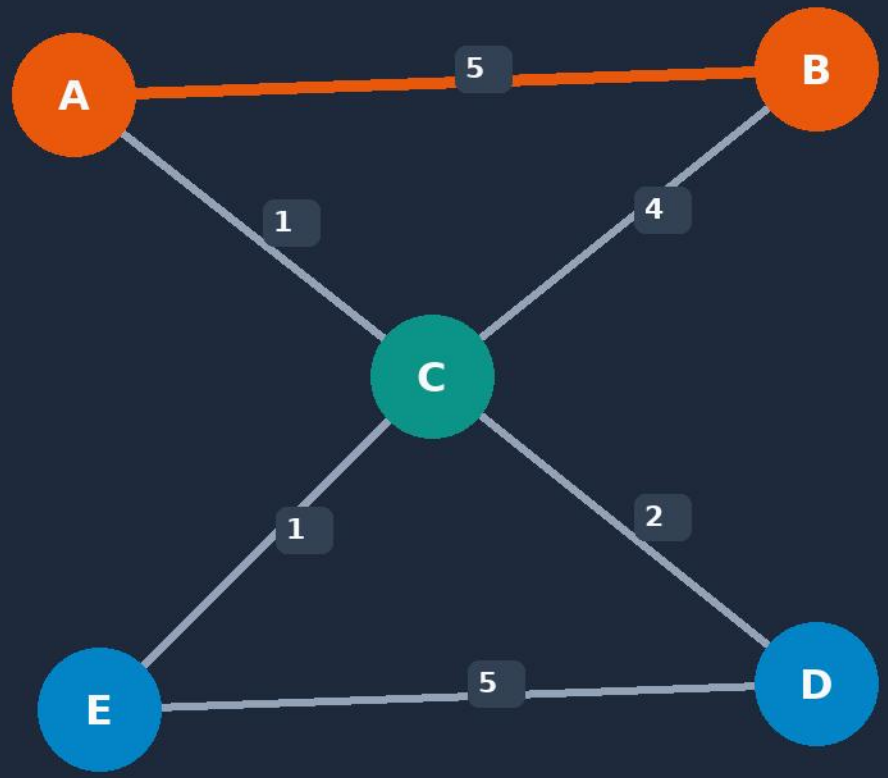


Walkthrough

- Begin with 5 singletons.
- Target K equals 2.
- All-separate cutsize is 18.
- Heaviest edge is A–B at weight 5.

Step 2: Merge 1 (Contract A-B)

Clusters: AB | C | D | E | Cutsize: 13

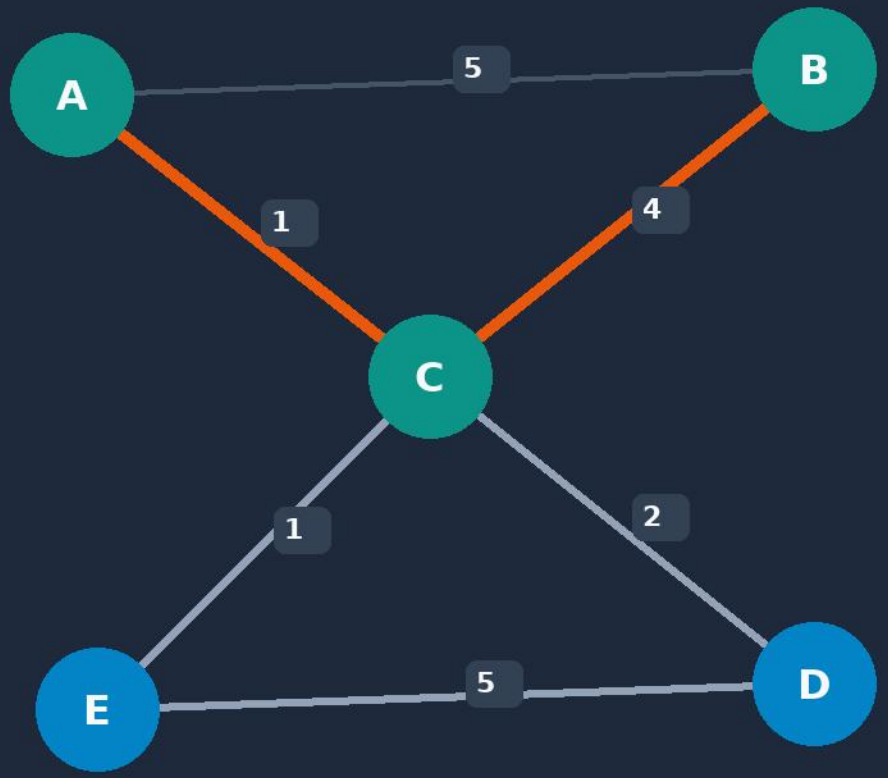


Walkthrough

- Merge A–B at weight 5.
- Protect the strongest connection first.
- Recompute edges into supernode AB.
- Clusters: AB | C | D | E, cutsize 13.

Step 3: Merge 2 (Absorb C into AB)

Clusters: ABC | D | E | Cutsize: 8

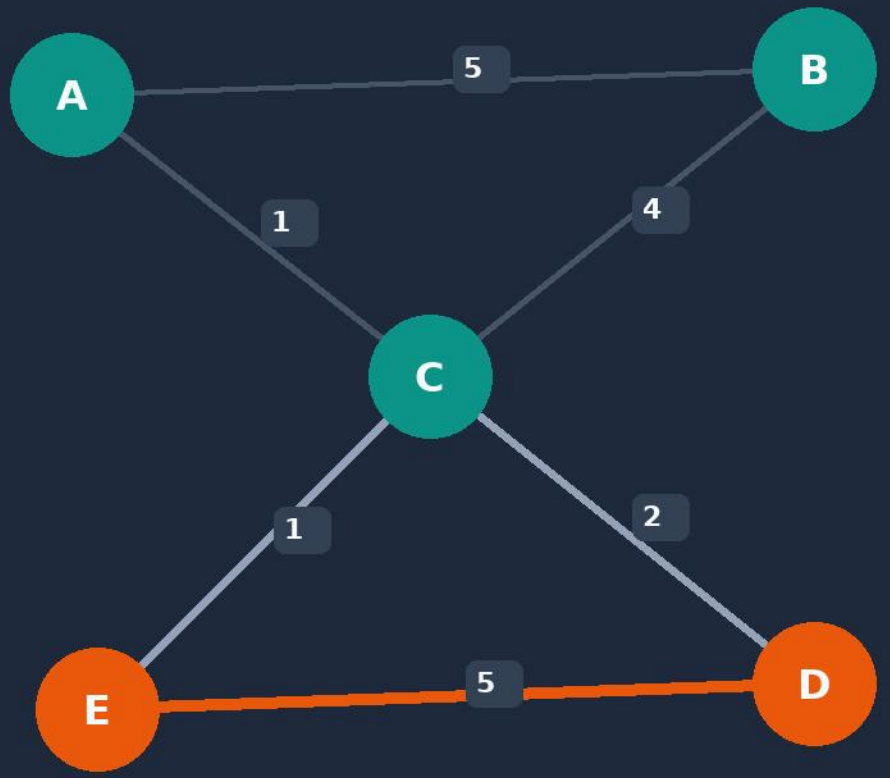


Walkthrough

- C–AB affinity is 1 + 4 = 5.
- That ties D–E at weight 5.
- Lexicographic rule: AB < D, so C merges.
- Clusters: ABC | D | E, cutsize 8.

Step 4: Merge 3 (Contract D-E)

Clusters: ABC | DE | Cutsize: 3 (Target K=2 Reached)

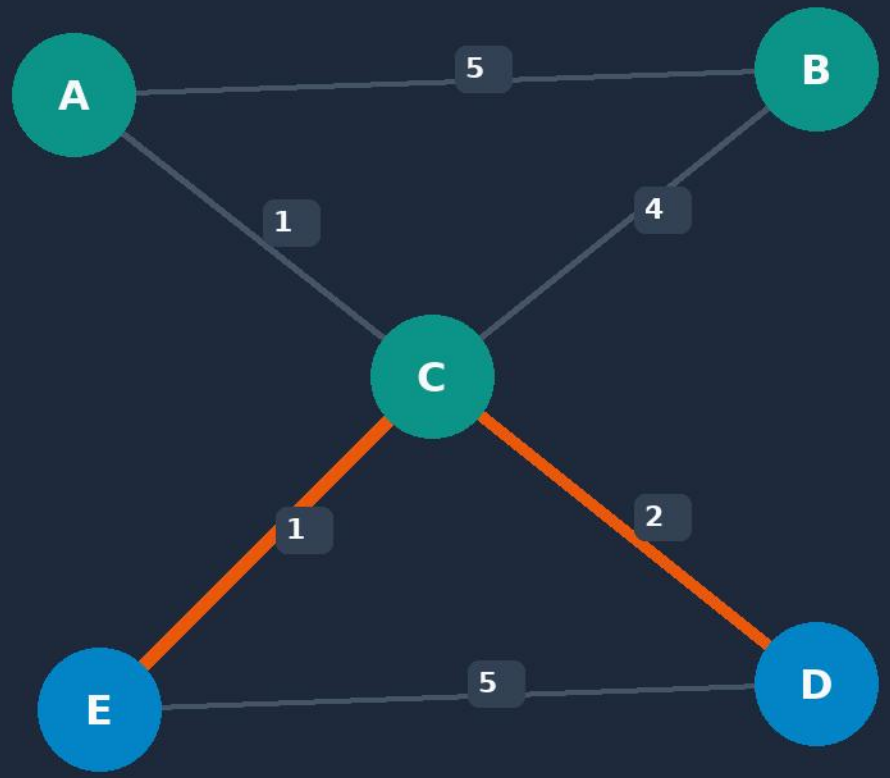


Walkthrough

- Last merge: D–E at weight 5.
- Two clusters: ABC versus DE.
- Target K=2 is reached.
- Only weak bridges remain on the cut.

Step 5: Final Result & Capacity Behavior

Final Partition: ABC | DE | Remaining Cutsize: 3



Walkthrough

- Cut edges: C–D weight 2, C–E weight 1.
- Unconstrained golden: ABC|DE, cut 3.
- Capacity=2 blocks size-3 ABC.
- Cap-2 result: AB|C|DE, cutsize 8.

Interactive Lab Challenges & Implementation

Key Insights & Workflow Pitfalls

- C1: C–AB fires over D–E due to tie-break (AB < D).
- C2: Capacity=2 strands the run at 3 clusters (AB|C|DE).
- C3: Golden cut edges are C–D (2) and C–E (1).
- Pitfall: test capacity before ranking; keep a stable tie-break.

Summary: Greedy Pair-Merge

Key Takeaways for CAD & Graph Partitioning

- Fast bottom-up coarsening engine for CAD and mesh problems.
- No backtracking: a local optimum may not be global.
- Capacity limits can prevent reaching target cluster count K.
- Correct affinity updates and stable tie-breaks are required.